

Designing a Mixed Reality Library Experience to Engage Primary Students with Open-Source Literacy Content

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Figure 1: Hardware and Software Prototypes for Immersive Mixed Reality Displays Library Interface for 3rd Grade Classrooms.

ABSTRACT

Highly immersive, interactive VR is engaging and effective for education and training across users, contexts, and domains. However, VR tools are not recommended for early elementary school children. Here we report on the participatory design approach to three novel mixed reality displays and a mixed reality software, “The Readerverse,” in the context of reading proficiency to demonstrate application of the affordances of mixed reality for early elementary education. This mixed reality library allows students to open virtual books that transport them into the book world with content specific learning experiences and games. This paper discusses pilot testing and iterative design with stakeholders.

Index terms: Mixed Reality, Design, Educational Technology, Audio Visual Systems, Human-Centered Computing

INTRODUCTION

Only 32 percent of US fourth graders are reading at or above grade level [1]. Despite growing use of digital strategies in education [2], student reading has not significantly improved since 1992 [1]. Third grade reading level is a significant predictor of eighth grade reading level, and subsequently adult literacy [2]. This manifests in adulthood, as 31% of 2019-20 US first-year college students took at least one remedial course in reading/writing or math [3]. The core issue lies in the prescriptive nature of programming, which often fails to engage children's interests, a key component in fostering reading motivation [4]. The demonstrated effectiveness of

highly immersive interactive augmented and mixed reality for engaging learners [5] along with risks of VR for children under 12, has led our team to design highly immersive interactive mixed reality interfaces to engage primary students with personalized literacy experiences in adaptive mixed reality library ecosystem.

BACKGROUND AND RELATED WORK

Immersive tools have been identified as having the potential to increase reading comprehension, intrinsic motivation, transfer [6], presence, and collaboration [7], [8]. They have also been identified as increasing the cognitive load, causing eye-strain, visual distortion and headaches [9]. Fortunately, some of the limitations to reading in VR are mitigated by mixed reality experiences, as these experiences may not require a user to wear a headset [11]. The research aim is to identify, apply, and test the design features that would make students most likely to engage with literacy.

Levels of Immersion

Immersion increases as the user is able to engage more naturally in an environment, as with mixed reality experiences that allow the user to engage with real and virtual object simultaneously at increasing levels. Highly immersive, interactive mixed reality experiences allow a user to engage fully with virtual objects in ways that seamlessly blend with the physical world [7]. Likewise, students have been shown to be more motivated to read when learning was augmented with a relevant immersive game [13].

THINK TANK FOCUS GROUP

To begin to address the literacy problem in third grade students, we gathered a team of subject matter experts across disciplines. The consolidated areas of expertise included literacy education, computer science, design thinking, k12 education, primary school teachers, and primary school teacher educator, virtual reality design and development, accessibility, and program evaluation.

The focus group began the process by cataloguing the barriers to reading proficiency identified in the literature. The most salient of these were lack of interest, self-regulation, self-efficacy, and task perception [12]. Students who struggle with reading also lack strategies for phonemic awareness, vocabulary, fluency, phonics, and comprehension [13]. MR's capacity to engage the senses increases the student's sense of presence and immersion in the environment. Similar to immersive VR, immersion and interactivity in MR contribute to user interest through the sense of

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agency and embodied cognition afforded [14], [15]. A fundamental limitation includes lack of accessibility of many products [16].

Designing with Affordances & Limitations in Mind

Our think tank created three prototypes to engage the affordances of highly immersive interactive MR in a classroom setting. Our goal to create a mixed reality tool that students could fully engage with at school but also engage with at home led us to design a single software prototype and three experimental hardware prototypes.

The preliminary hardware designs can be seen in figure 1. In prototype one, the display is a large curved clear screen so they can see their teacher until the teacher activates the system. The large screen increases immersion by increasing the sense of presence. The quarter dome display in prototype two increases the level of immersion in virtual learning spaces, without requiring the user to wear a head-mounted display. Similar to traditional CAVE environments, this display will continue to allow the user to engage using their body as a controller in the world, but allows learners a more natural and realistic experiences of looking around a space. Prototype three includes the immersive interactive elements of the other MR designs, camera for gesture input, microphone for audio input, but adds more interactivity with a haptic mat that allows the phonics learner to engage with the environment and virtual humans. The haptic mat allows user to input commands and interact with the environment by stepping on the mat as directed on the screen. This mat will also add the capacity for the user to engage with the learning content through phonetic sounds, rhythm, and music.

Our immersive interactive library experience employs a pedagogical agent to explore and read with the learner (figure 1. Screen capture on the right). As students explore, a library of open-source books and audio from project Gutenberg and LibriVox are used to create a read along experience. Mixed reality elements employed are tracking user movement to simulate movement in virtual space. Natural language processing of student speech also allows natural interaction with texts and the pedagogical agent.

User Feedback on Prototypes

After our core team identified our most immersive interactive designs, we sought feedback on both the hardware and software prototypes from third grade teachers (N=15). The feedback was gathered by sending videos of each prototype to primary school teachers around the United States. After completing informed consent, teachers rated and commented on each prototype.

Feedback from these teachers suggest that the clear curved screen and haptic mat were perceived as most beneficial in the classroom. Teachers appreciated the capacity for touch, immersion, and movement while maintaining personal space boundaries.

To test our software prototype, we set up a demonstration in a local museum and let children try the pilot experience. After we attained their consent to collect data after the experience users explored the environment. We then conducted semi-structured interviews with children (N=15) and accompanying adults (N= 20).

An initial thematic analysis of the interviews and interactions revealed that one quarter of students started to lose interest after seven or eight pages (looking around the room and ultimately leaving). About half of the participants noted that they had used something similar to this system in school (Epic), except that “it wasn’t as fun” or “you couldn’t move around in it.”

CONCLUSION

Our three innovative prototypes—a curved clear display, a quarter dome display, and a display with a haptic mat—each uniquely leverage mixed reality's affordances to enhance student engagement and learning outcomes. While we are in the

preliminary design stages, it was very encouraging to find that teachers were very enthusiastic about the prototypes. Likewise, the positive response to the software was encouraging while providing guidance for the next steps to increase usability and engagement.

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